

# Tianyi Zhang

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## EDUCATION

### Georgia Institute of Technology

M.S. in Computer Science

Atlanta, GA

Aug 2025 – May 2027

### Emory University

B.S. in Quantitative Sciences (Data Science)

Atlanta, GA

Aug 2021 – May 2025

## PROFESSIONAL EXPERIENCE

### LinkedIn Corporation

Machine Learning Engineer Intern

Mountain View, CA

May 2026 – Aug 2026

- Corrected data and loss implementation issues in a production **two-tower retriever**; audited dwell-heavy labels, missing rejection supervision, and imputed negatives to distinguish training-signal gaps from evaluation artifacts.
- Designed and handed off a **Qwen multi-tower architecture** with separate profile/history member encoders and learnable routing; defined candidate-complementarity gates under a fixed ANN budget to preserve distinct interests without expanding ranker load.
- Designed a serving-aware **VLM-teacher / embedding-student path**: selectively ground metadata, text, and images into evidence for soft relevance labels and hard-negative mining, then precompute compact candidate vectors for ANN retrieval.
- Shipped a **PySpark evaluation framework over 2.7B member-candidate pairs**, measuring relevance, coverage, and topical diversity across matched lifecycle/signal cohorts; found Semi-cold members had **25% higher Recall@10 but 22% lower ILD@10** than Active members, informing signal-validation and diversity gates.

### Georgia Tech EIC Lab · NVIDIA AI Research

Student Researcher

Atlanta, GA

Jul 2025 – Dec 2025

- Engineered the **AgentSuite** benchmark-auditing pipeline with a taxonomy-guided LLM judge and deterministic structural detectors, matching expert audits at **0.79–0.87 F1** across **6 benchmarks**.
- Generalized to unseen benchmarks without retuning; identified flaws in **23% of tasks** whose correction changed **63% of model rankings**.

### Emory Graph Mining Lab

Machine Learning Researcher

Atlanta, GA

Dec 2023 – May 2025

- Designed a **Granger-causal DAG framework** that converts fMRI time series into directed brain networks, exposing asymmetric dependencies hidden by correlation graphs and improving diagnostic prediction.
- Co-developed a **Brain Network State Transformer** that clusters dynamic connectivity into recurring latent states and models intra-/inter-state structure for downstream prediction.

## OPEN SOURCE

### NVIDIA NeMo RL (2.0K stars)

Merged PRs  Open PRs 

Open-Source Contributor | Scalable toolkit for efficient LLM reinforcement learning and post-training.

- Performance & Correctness (merged)**: Optimized log-probability and top- $k$  distillation paths, making cross-token P-KL **3.7× faster** and reducing tensor-parallel communication from **250 MiB to 16 MiB per rank**.
- Multi-Reward GDPO (under review)**: Preserved named reward components end to end across async rollout, replay, checkpoint/restore, and custom policy objectives, enabling composable multi-reward post-training.
- SingleController PPO (under review)**: Streamed actor and critic minibatches instead of materializing merged payloads, reducing memory pressure while preserving functional resume and training parity on H100s.

## PUBLICATIONS

H. Suh\*, S. Lee\*, B. Ji\*, R. Khare, B. Khan, H. Kim, **T. Zhang**, et al. “AgentSuite: Toward More Reliable Agent Evaluation with a Component-Based Benchmark Auditing Pipeline” *International Conference on Machine Learning (ICML)* 2026.

**T. Zhang**, K. Han, J. Nie, et al. “Causal Brain Connectivity: Integrating Granger Directed Graphs in fMRI Analysis.” *International Conference on Artificial Intelligence in Medicine (AIME)* 2025.

J. Nie, K. Han, **T. Zhang**, et al. “Brain Network State Transformer: Leveraging State Functional Connectivity.” *IEEE Engineering in Medicine and Biology Society (EMBC)* 2025. (Oral)

## TECHNICAL SKILLS

|               |                                                                                   |
|---------------|-----------------------------------------------------------------------------------|
| Languages     | Python, Go, SQL, C/C++, Java, R                                                   |
| ML Frameworks | PyTorch, TensorFlow, PyTorch Lightning, Hugging Face Transformers                 |
| Post-Training | NVIDIA NeMo RL, verl, vLLM, SGLang, CUDA                                          |
| Data & Tools  | PySpark, Apache Spark, Flyte, MLflow, FastAPI, Docker, AWS, GitHub Actions, Linux |